

OSI and TCP/IP Model Notes

OSI Model

The OSI (Open Systems Interconnection) Model is a seven-layer conceptual framework developed by ISO to standardize network communication.

Seven Layers of OSI Model:

1. Application Layer – Provides network services to end-user applications.
2. Presentation Layer – Translates, encrypts, and compresses data.
3. Session Layer – Manages and controls communication sessions.
4. Transport Layer – Ensures reliable end-to-end data delivery.
5. Network Layer – Handles logical addressing and routing.
6. Data Link Layer – Manages MAC addressing and error detection.
7. Physical Layer – Transmits raw bits over physical media.

TCP/IP Model

The TCP/IP (Transmission Control Protocol/Internet Protocol) Model is a four-layer networking framework developed by DARPA and used on the Internet.

Four Layers of TCP/IP Model:

1. Application Layer – Provides services to user applications.
2. Transport Layer – Ensures end-to-end communication (TCP/UDP).
3. Internet Layer – Manages IP addressing and routing.
4. Network Access Layer – Handles physical transmission and MAC addressing.